UNIVERSITY OF BIRMINGHAM DUBAI

برنامج الذكاء الاصطناعي
THE ARTIFICIAL INTELLIGENCE PROGRAM

AERO UAE
AUTONOMOUS DRONE DELIVERY

The Autonomous Drone Delivery Agent

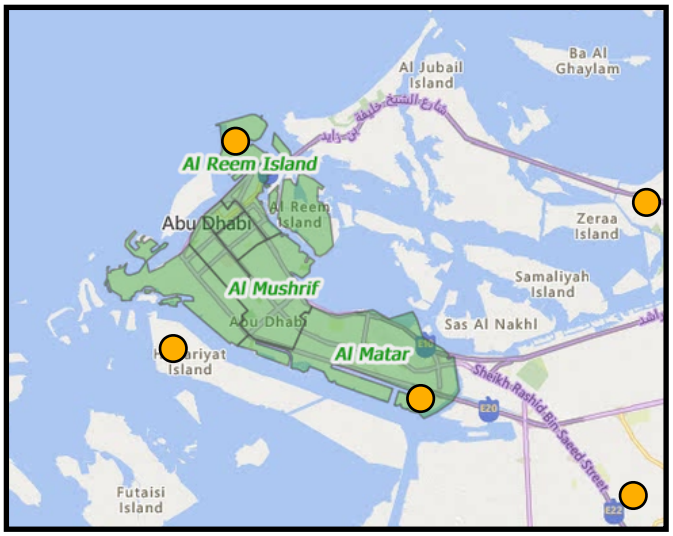
What information (data) does an AI agent need to coordinate a fleet of delivery drones across a city?

The Mission

You are developing an AI agent to organise a fleet of 500 **autonomous delivery drones** across 5 delivery points in Abu Dhabi.

Your agent is responsible for:
- Assigning delivery requests to available drones
- Choosing routes and estimating delivery times
- Monitoring ongoing deliveries
- Deciding when to reroute, delay, or reassign a delivery
- Learning from previous operations to improve future decisions

It receives delivery requests from each zone and must organise how the movement of the drones.



Section 1: What does the agent need to know?

This section is focused on thinking about what information your agent needs to operate on. For now, do not think about *preprocessing*, but just what *data sources* your agent needs.
For each data item, mention:
- **where** this information will come from
- **how** often it must be updated
- **why** the agent needs this data

*Think about what data your agent needs from each **drone**. What information will the agent need and **what sensors** would you want on the drones?*

*Think about what data your agent needs from each **delivery point**. What information do you need and how would you **collect it** at the delivery point?*

*Think about what data your agent needs from each **route** (the paths from delivery point to delivery point). What information do you need and how would you **collect it**?*

Section 2: From Raw Data to Actionable Information (Data Preprocessing and Representation)

Your agent now has access to the data sources you identified in Section 1 related to the drones, delivery points, and routes. However, raw data is not necessarily ready for an AI agent to use. Some observations may be incomplete, noisy, duplicated, outdated, recorded at different frequencies, or too large to process efficiently.

Design a preprocessing pipeline that transforms your collected data into reliable, compact, and useful information for the agent's decisions.

Consider the data sources you identified in Section 1. For three different data sources, answer the following questions:

1. What **data-quality problems** might you encounter? Consider missing values, noisy measurements, incorrect records, duplicates, and outliers.
2. What **preprocessing techniques** would you apply to address these problems? Be specific: would you filter noisy measurements, interpolate missing values, remove duplicates, standardize formats, or flag suspicious observations?
3. When might preprocessing introduce **errors**? ***Think carefully:*** Should all missing values be filled in? When is it better to preserve missing information and communicate uncertainty to the agent?

[Data Source 1] _____

[Data Source 2] _____

[Data Source 3] _____

If different sources need to be combined, how would you **align** them? For example, matching timestamps, geographical coordinates, or identifiers.

Some of your chosen sources may contain **large amounts of information**, such as drone camera footage, satellite imagery, detailed maps, or historical flight trajectories.

The agent cannot necessarily process or store all this information in its original form. *Identify one or two data sources that would be particularly expensive to store, transmit, or process.*

What information from these sources would actually be useful to the agent?

How would you transform the raw data into a smaller, structured representation? For example, could video footage be converted into geolocated obstacle records, or satellite imagery into a map of buildings and restricted areas?

What information might be lost through compression, aggregation, or feature extraction? Under what circumstances would you need to retain or revisit the original data?

Section 3: Memory and Data During Deployment

Your drone delivery network is now operational across Abu Dhabi's five zones. Every day, your 500 drones complete hundreds of deliveries, generating new observations about their own performance, delivery locations, and the environment.

Section 3.1 Designing your Agent's Memory

What information should your agent remember about individual drones, delivery points, and routes to make better decisions in the future?

Design **three types of memory**. Specify **what information each contains**, why it would help the agent **make better decisions over time**, how it is **updated**, and when observations should expire or be revalidated.

[Memory 1] _____

[Memory 2] _____

[Memory 3] _____